

GPT-5.5 frontier saturation check

Stories generated from Equational Theories Project implications (storyform.py), formalized back by each model over OpenRouter, and graded syntactically (checkform.py). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

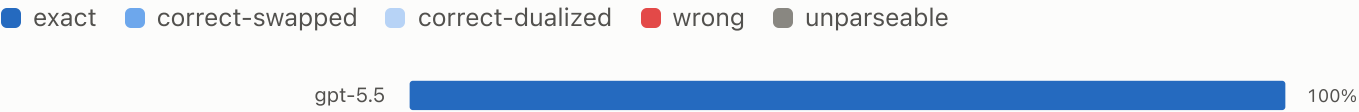
experiments/06-frontier-saturation/runs/run-s0-n20-story-gpt55-think-on · experiments/06-frontier-saturation/runs/run-strat5-s0-story-gpt55-think-on

Runs	Models	Stories	Graded calls	Overall correct
2	1	60	60	100%

20 uniformly sampled pairs · seed 0 · all-think

FIGURE 1 Verdict composition — all-think regime · uniform 20

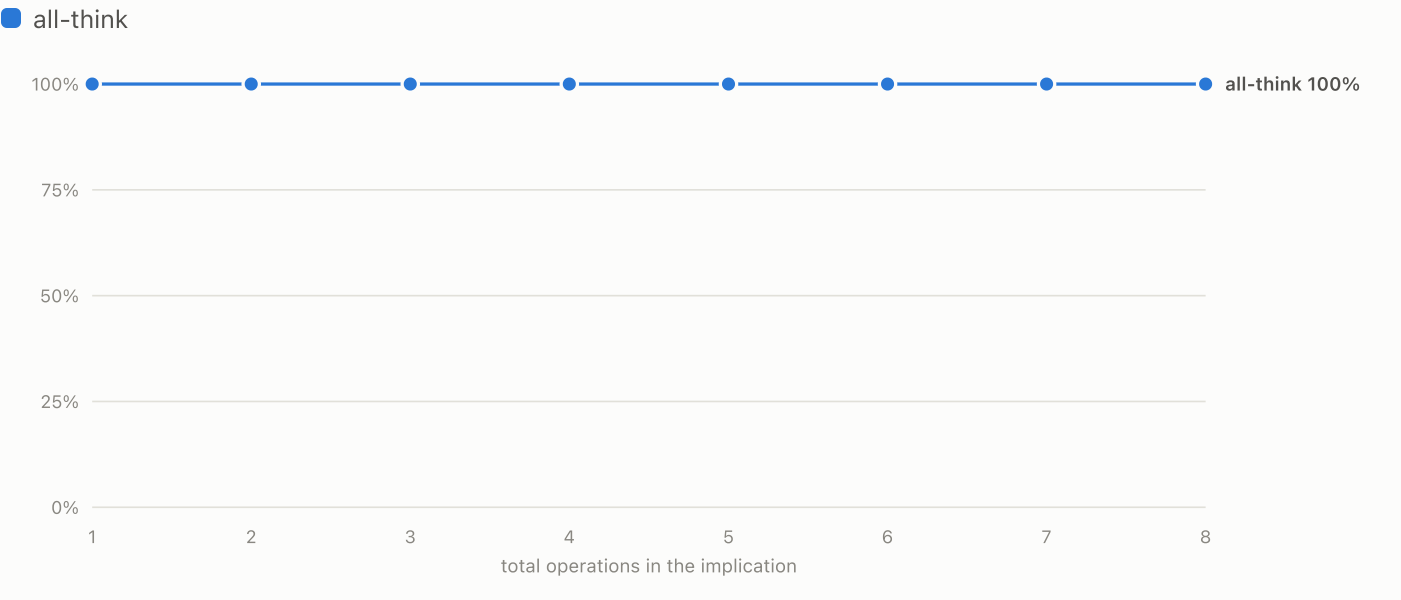
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 20 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.



► Data: verdict counts per model

FIGURE 2 Accuracy vs operation count — stratified 40

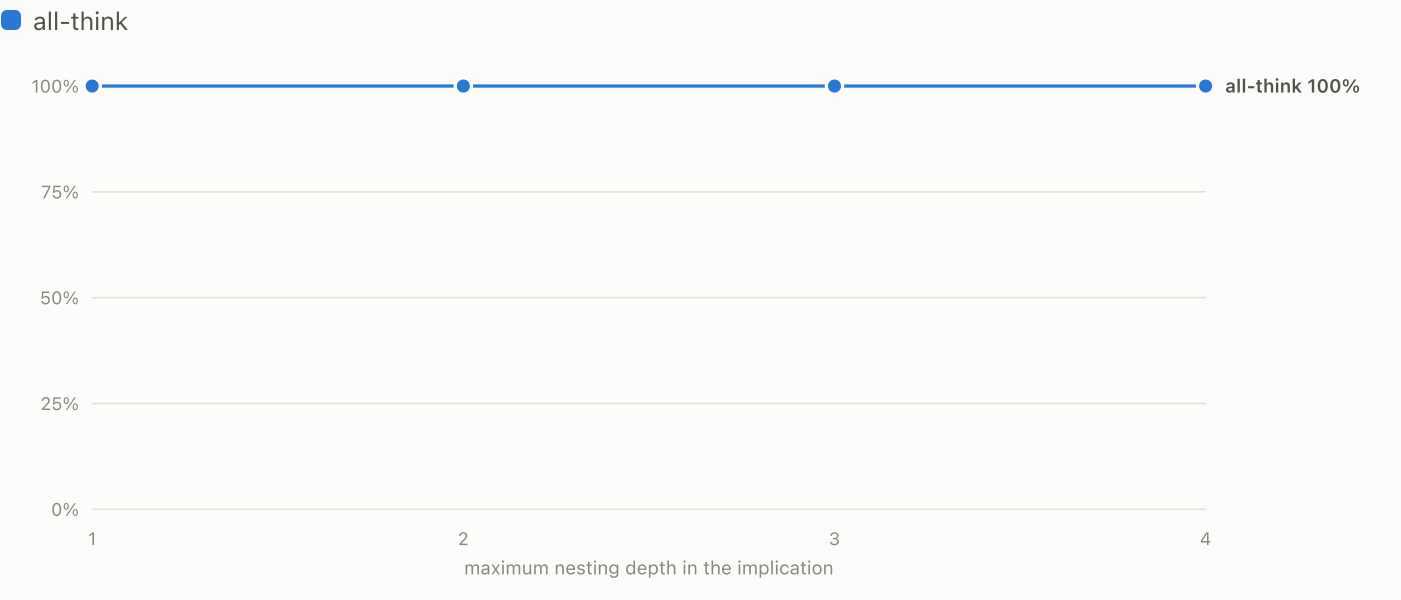
Correct rate pooled over all models, by the implication's total operation count (both equations combined). Sample: 40 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: pooled correct rate per operation-count bin

FIGURE 3 Accuracy vs nesting depth — stratified 40

Correct rate pooled over all models, by the deepest operation nesting across the four equation sides. Sample: 40 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: pooled correct rate per depth bin

FIGURE 4 Verdict composition — all-think regime · stratified 40

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

Sample: 40 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

gpt-5.5  100%

► Data: verdict counts per model